Housing Price Prediction using Multiple Linear Regression

1. Introduction:

The housing market plays a crucial role in the economy, with property prices significantly influencing investment decisions. As real estate data continues to grow, analyzing the factors that drive housing prices becomes increasingly important. This project aims to use multiple linear regression (MLR) to predict housing prices in King County, Washington, based on various property features. By analyzing a dataset from Kaggle, this project will identify key factors influencing high property values and develop a predictive model to aid real estate decision-making.

2. Objectives:

- Investigate patterns between property features (e.g., size, bedrooms, condition) and housing prices.
- Identify key factors influencing high-value properties (over $650K).
- Build a predictive model using MLR to forecast housing prices.
- Provide actionable insights for real estate professionals to enhance pricing strategies.

3. Data Collection:

The dataset is sourced from Kaggle and contains 21 variables with 21,613 observations, including features such as square footage, number of bedrooms, and property condition. The data covers house sales in King County from May 2014 to May 2015.

4. Methodology:

- **Data Cleaning & EDA**: Select important features based on correlation, and perform outlier removal to enhance model accuracy.
- **Data Preprocessing**: Split the data into training and testing sets and create dummy variables for categorical features.
- **Modeling**: Use multiple linear regression to predict housing prices and evaluate model performance using R^2, coefficients, and p-values.

5. Deliverables:

- A Jupyter notebook containing:
  - Detailed report with findings and insights.
  - Visualizations of key features and trends.
  - Model evaluation metrics and code for data preprocessing, feature engineering, and modeling.

6. Conclusion:

This project will provide valuable insights into housing price determinants in King County, offering a robust predictive model for real estate pricing decisions.